Ray tracing – Curved quadratic elements

1. Tetrahedron and ray

Surface position on the tetrahedron's face (weighted sum of shape functions):

$$\boldsymbol{P}(g,h) = \sum_{i=0}^{5} N_i(g,h) \cdot \boldsymbol{f_{nodes_i}} \tag{1}$$

Ray:

$$\boldsymbol{R}(t) = \boldsymbol{o} + t\boldsymbol{d} \tag{2}$$

where $\boldsymbol{d}$ is the ray's direction, and $\boldsymbol{o}$ is the ray's origin.

2. Intersection by minimising vector difference

2.1. Intersection problem

Face-ray intersection (3 unknowns) presented as the vector difference [1, 2]:

$$\boldsymbol{o} + t\boldsymbol{d} - P(g,h) = 0 \tag{3}$$

Nonlinear system of equations:

$$\boldsymbol{F}(t,g,h) = \boldsymbol{o} + t\boldsymbol{d} - \sum_{i-0}^{5} N_i(g,h) \cdot f_{nodes_i} = \boldsymbol{0} \tag{4}$$

The system of equations contains 3 unknowns and 3 equations, so the mapping is:

$$\Re^3 \rightarrow \Re^3 \tag{5}$$

They represent the distance vector between a point on the ray and a point on the surface.

2.2. Möller-Trumbore algorithm for linear triangle intersection (initial guess)

$$\boldsymbol{E_1} = V_1 - V_0,\ \boldsymbol{E_2} = V_2 - V_0 \tag{6}$$
$$\boldsymbol{p} = \boldsymbol{d} \times \boldsymbol{E_2} \tag{7}$$
$$det = \boldsymbol{E_1} \cdot \boldsymbol{p} \tag{8}$$
$$\boldsymbol{t_{vec}} = \boldsymbol{o} - V_0 \tag{9}$$

Where V_0, V_1, V_2 are triangle vertices.

$$u = \frac{\boldsymbol{t_{vec}} \cdot \boldsymbol{p}}{det} \tag{10}$$

$$\boldsymbol{q} = \boldsymbol{t_{vec}} \times \boldsymbol{E_1} \tag{11}$$

$$v = \frac{\boldsymbol{d} \cdot \boldsymbol{q}}{det} \tag{12}$$

$$t = \frac{\boldsymbol{E_2} \cdot \boldsymbol{q}}{det} \tag{13}$$

2.3. Newton-Raphson method

Jacobian of F:

$$\boldsymbol{J_F} = \left[\frac{\partial F}{\partial f}, \frac{\partial F}{\partial g}, \frac{\partial F}{\partial h}\right] \tag{14}$$
$$\boldsymbol{J_F} = \left[\boldsymbol{d}, -\frac{\partial P}{\partial g}, -\frac{\partial P}{\partial h}\right] \tag{15}$$

$$\frac{\partial \boldsymbol{P}}{\partial g} = \sum_{i=0}^{5} \frac{\partial N_i}{\partial g} \cdot \boldsymbol{f_{nodes_i}} \tag{16}$$
$$\frac{\partial \boldsymbol{P}}{\partial h} = \sum_{i=0}^{5} \frac{\partial N_i}{\partial h} \cdot \boldsymbol{f_{nodes_i}} \tag{17}$$

The update by solving a linear system:

$$\boldsymbol{\Delta} = [\Delta t, \Delta g, \Delta h]^T \tag{18}$$

$$\boldsymbol{J_F} \cdot \boldsymbol{\Delta} = -F(t, g, h) \tag{19}$$

$$\begin{bmatrix} \boldsymbol{d}_x & -\frac{\partial P_x}{\partial g} & -\frac{\partial P_x}{\partial h} \\ \boldsymbol{d}_y & -\frac{\partial P_y}{\partial g} & -\frac{\partial P_y}{\partial h} \\ \boldsymbol{d}_z & -\frac{\partial P_z}{\partial g} & -\frac{\partial P_z}{\partial h} \end{bmatrix} \begin{bmatrix} \Delta t \\ \Delta g \\ \Delta h \end{bmatrix} = -(\boldsymbol{o} + t\boldsymbol{d} - P(g, h)) \tag{20}$$

3. Intersection by minimising scalar squared distance

3.1. Intersection problem

Face-ray intersection could be presented as the scalar squared difference [3]:

$$F(g, h) = |\boldsymbol{V}|^2 - (\boldsymbol{V} \cdot \boldsymbol{d})^2 \tag{21}$$

where

$$\boldsymbol{V} = \boldsymbol{P}(g, h) - \boldsymbol{o} \tag{22}$$

and

$$t = \boldsymbol{V} \cdot \boldsymbol{d} = (\boldsymbol{P}(g, h) - \boldsymbol{o}) \cdot \boldsymbol{d} \tag{23}$$

The equation contains 2 unknowns, so the mapping is:

$$\Re^2 \to \Re^1 \tag{24}$$

They represent the squared perpendicular distance between the ray and surface (Figure 1).

If a local minimum of F is zero, then this corresponds to a point where the ray intersects the surface. Those points where F is a minimum, but where $F > 0$, indicate that the ray misses the surface by a finite distance. This is actually desirable in that the algorithm will still converge near "silhouette edges" of surface, but will give a non-zero minimum.

F could be split into 2 functions:

$$\boldsymbol{F}(g, h) = |\boldsymbol{V}|^2 - (\boldsymbol{V} \cdot \boldsymbol{d})^2 \tag{25}$$

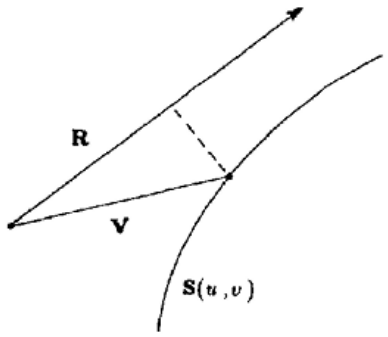

Figure 1: Distance from a point on the surface to a ray.

3.2. Quasi-Newton method

Jacobian of F:

$$\boldsymbol{J_F} = \left[\frac{\partial F}{\partial g}, \frac{\partial F}{\partial h}\right]^T \tag{26}$$

The partials of F are:

$$\frac{\partial F}{\partial g}(g,h) = 2\left(\frac{\partial \boldsymbol{P}}{\partial g}\cdot\boldsymbol{V} - t\frac{\partial \boldsymbol{P}}{\partial g}\cdot\boldsymbol{d}\right) = \tag{27}$$

$$= 2\frac{\partial \boldsymbol{P}}{\partial g}\left(\boldsymbol{V} - t\boldsymbol{d}\right) = \tag{28}$$

$$= 2\frac{\partial \boldsymbol{P}}{\partial g}\left(\boldsymbol{P} - (\boldsymbol{o} + t\boldsymbol{d})\right) \tag{29}$$

$$\frac{\partial F}{\partial h}(g,h) = 2\frac{\partial \boldsymbol{P}}{\partial h}\left(\boldsymbol{P} - (\boldsymbol{o} + t\boldsymbol{d})\right) \tag{30}$$

Hessian of F:

$$\boldsymbol{H_F} = \begin{bmatrix} \frac{\partial^2 F}{\partial g^2} & \frac{\partial^2 F}{\partial gh} \\ \frac{\partial^2 F}{\partial hg} & \frac{\partial^2 F}{\partial h^2} \end{bmatrix} \tag{31}$$

where

$$\frac{\partial F}{\partial g^2}(g,h) = 2\frac{\partial^2 \boldsymbol{P}}{\partial g^2}\left(\boldsymbol{P} + 1 - \boldsymbol{R}\right) \tag{32}$$

$$\frac{\partial F}{\partial h^2}(g,h) = 2\frac{\partial^2 \boldsymbol{P}}{\partial h^2}\left(\boldsymbol{P} + 1 - \boldsymbol{R}\right) \tag{33}$$

$$\frac{\partial F}{\partial gh}(g,h) = \frac{\partial F}{\partial hg}(g,h) = 2\frac{\partial^2 \boldsymbol{P}}{\partial gh}\left(\boldsymbol{P} + 1 - \boldsymbol{R}\right) \tag{34}$$

The update by solving a linear system:

$$\boldsymbol{\Delta} = [\Delta g, \Delta h]^T \tag{35}$$

$$\boldsymbol{H_F} \cdot \boldsymbol{\Delta} = -\boldsymbol{J_F} \tag{36}$$

$$\begin{bmatrix} \frac{\partial^2 F}{\partial g^2} & \frac{\partial^2 F}{\partial gh} \\ \frac{\partial^2 F}{\partial hg} & \frac{\partial^2 F}{\partial h^2} \end{bmatrix} \begin{bmatrix} \Delta g \\ \Delta h \end{bmatrix} = -\left[\frac{\partial F}{\partial g}, \frac{\partial F}{\partial h}\right]^T \tag{37}$$

4. Surface normal

$$\boldsymbol{n} = \frac{\partial \boldsymbol{P}}{\partial g} \times \frac{\partial \boldsymbol{P}}{\partial h} \tag{38}$$

References

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[2] D. L. Toth, On ray tracing parametric surfaces, SIGGRAPH Comput. Graph. 19 (1985) 171–179. URL: https://doi.org/10.1145/325165.325233. doi:10.1145/325165.325233.

[3] K. I. Joy, M. N. Bhetanabhotla, Ray tracing parametric surface patches utilizing numerical techniques and ray coherence, SIGGRAPH Comput. Graph. 20 (1986) 279–285. URL: https://doi.org/10.1145/15886.15917. doi:10.1145/15886.15917.